

26TH DIGITAL BLUE ECONOMY SUMMIT · EXPERIMENTAL PAPER

ParaSail

Predictive Advisory and Retrieval-Augmented System Advancing Informed Livelihoods

A geospatial decision support system for sustainable fishing, marine conservation and coastal livelihood resilience

Hard conservation constraints

Explainable by retrieval

Adaptable by LoRA fine-tuning

Grounded AI assistant (local)

Anantha Krishnan AS · Naipunnaya School of Management, Cherthala, India · September 2026

The blue economy under pressure

35.5%

of assessed stocks fished beyond sustainable levels

FAO, State of World Fisheries 2023

11 Mt

of plastic entering the ocean every year

UNEP, Turning off the Tap 2023

~3×

projected growth of plastic flows by 2040

UNEP, Turning off the Tap 2023

~492 M

people supported by small-scale fisheries

FAO, Illuminating Hidden Harvests 2023

Rebuilding stocks under sound management pays; what is missing is the real-time information infrastructure for coastal actors to act on it.

Four harms of planning fishing blind

01

Economic

Fuel, time and labour wasted on blind search for shifting fish distributions.

02

Ecological

Fishing pressure concentrates on spawning grounds, sensitive habitats and protected zones.

03

Safety

Small vessels put to sea without granular marine wind and wave guidance.

04

Institutional

Weather apps, habitat studies, AIS analytics and regulations never speak to one another.

Existing tools are fragmented

Tool class	Weather	Fish context	Legality	Explainability
Weather applications	✓	—	—	—
PFZ-style advisories	—	✓	—	—
Habitat / SDM studies	—	✓	—	—
AIS fleet analytics	—	—	—	—
Static regulations	—	—	✓	—
ParaSail (this work)	✓	✓	✓	✓

No existing tool class fuses weather, fish context, legality and explanation — the gap is architectural, not algorithmic.

OBJECTIVES

What ParaSail sets out to do

“Recommend safe, legal and ecologically responsible fishing windows — with the justification attached.”

Eight objectives, each mapped to a validation test family (T1–T4).

RO1

Fuse live open data into one quality-controlled geospatial model

RO2

Model aggregation and movement tendency, with habitat fallback

RO3

Enforce MPAs and closures as hard constraints

RO4

Attach regulations, guidance and local knowledge to every advisory

RO5

Adapt locally via custom ingestion and LoRA fine-tuning

RO6

Expose a production-style API and visual dashboard

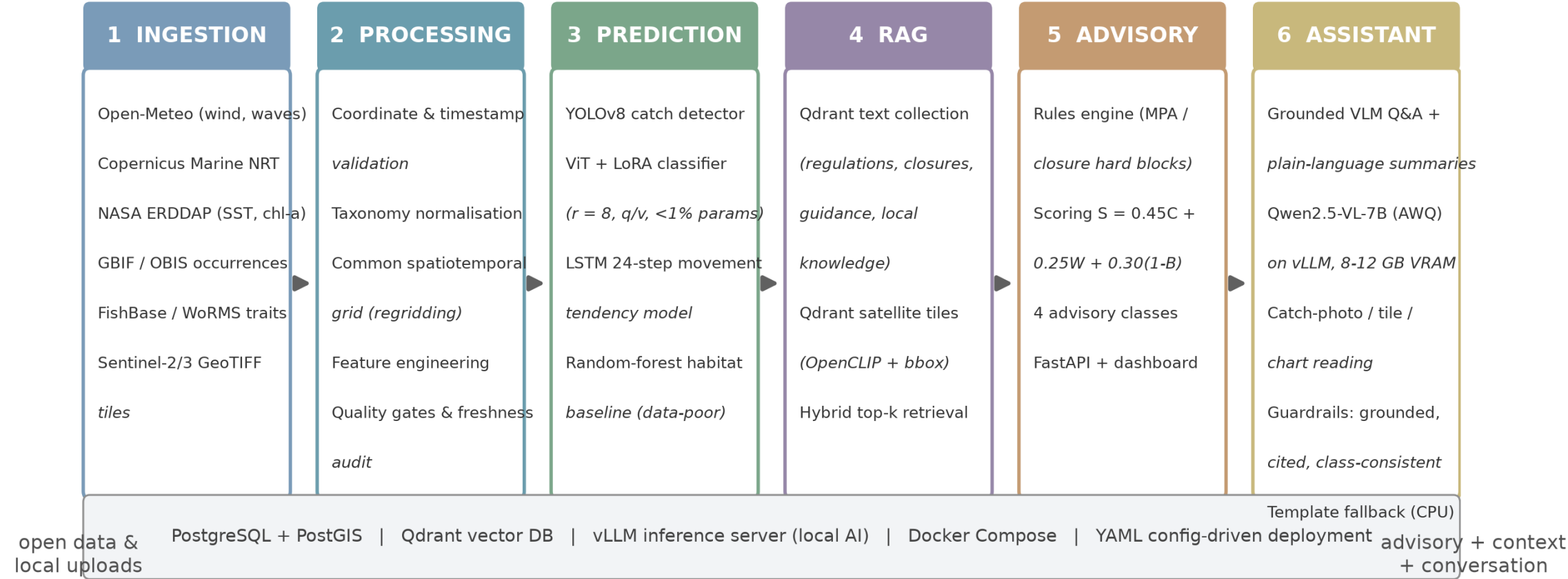
RO7

Evaluate via live-fetch, rule, scoring and retrieval tests

RO8

Align with blue-economy principles end to end

Six layers, one advisory



Every layer is independently replaceable and configured through YAML — region, species, closures, models and score weights are data, not code.

Live open data, normalised and audited

Source	Content	Cadence
Open-Meteo	Hourly meteorological and marine forecasts	Hourly
Copernicus Marine	Near-real-time ocean analysis (SST, currents)	Daily
NASA ERDDAP	SST and chlorophyll-a fields	Daily
GBIF / OBIS	Historical species occurrences	Corpus
FishBase / WoRMS	Taxonomy, traits, seasonality	Corpus
Sentinel-2 / Sentinel-3	Multispectral satellite tiles	Per pass

Resilience by design

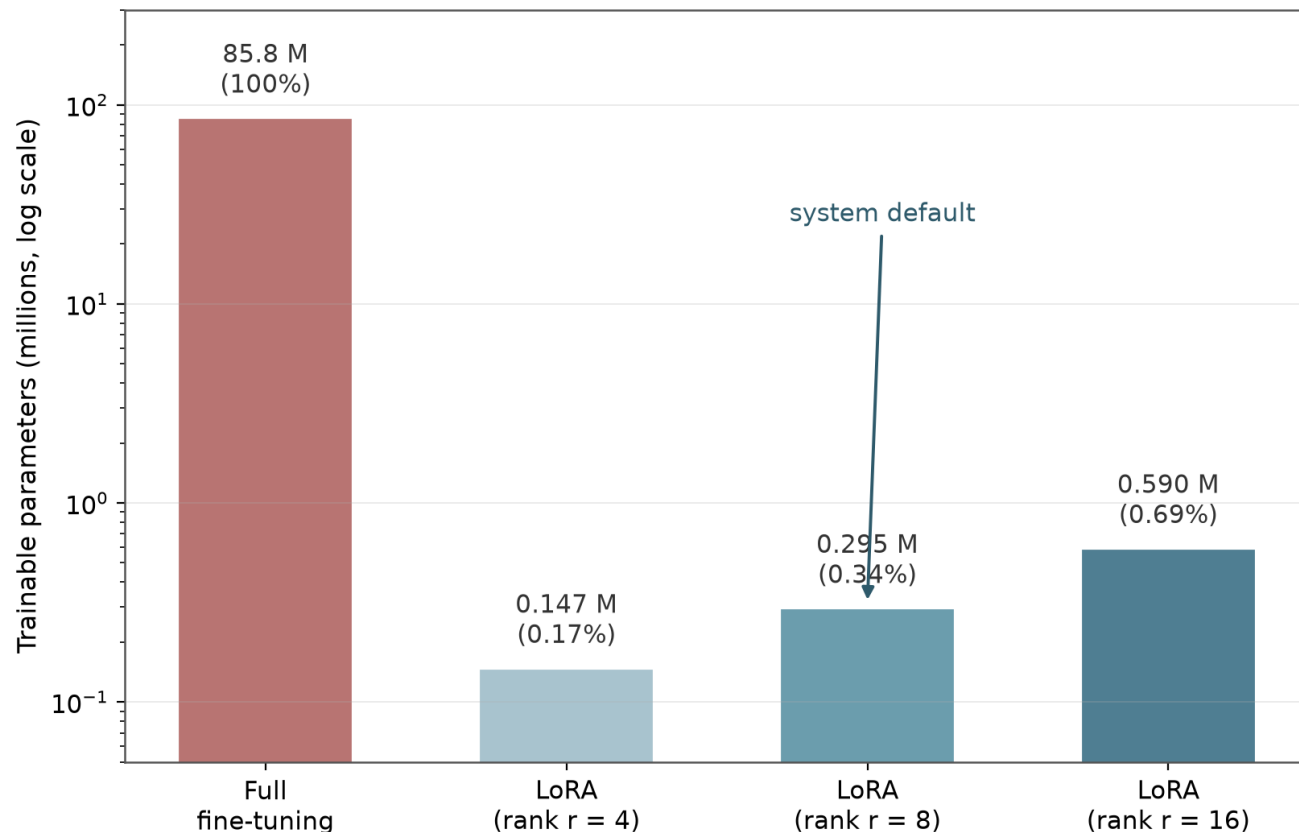
- Freshness contracts per source
- Response caching with age tracking
- Offline contingency: advisories continue from cached fields, with explicit age flags
- Quality gates and taxonomy normalisation (WoRMS identifiers)

Model selection: mixed families, deliberate trade-offs

Task	Selected model	Why this one
Catch-image detection	YOLOv8n (Ultralytics)	Real-time on modest hardware; mature toolchain
Species classification	ViT-B/16 + LoRA (r = 8)	Fine-grained accuracy at under 1% trainable parameters
Movement tendency (24 h)	Two-layer LSTM (hidden 128)	Short-horizon sequences for data-rich species
Habitat suitability	Random forest	Interpretable fallback for data-poor taxa
Text embeddings	all-MiniLM-L6-v2	Fast 384-dimensional sentence embeddings
Satellite tile embeddings	OpenCLIP ViT-B/32	Similarity search over unlabelled tiles
Assistant: summaries, Q&A, image reading	Qwen2.5-VL-7B (AWQ 4-bit) on vLLM	One open-source VLM for all assistant jobs; 8-12 GB VRAM; batched local serving — no per-token API costs

Selection criterion is coastal deployability, not leaderboard position — every model runs on mid-range hardware.

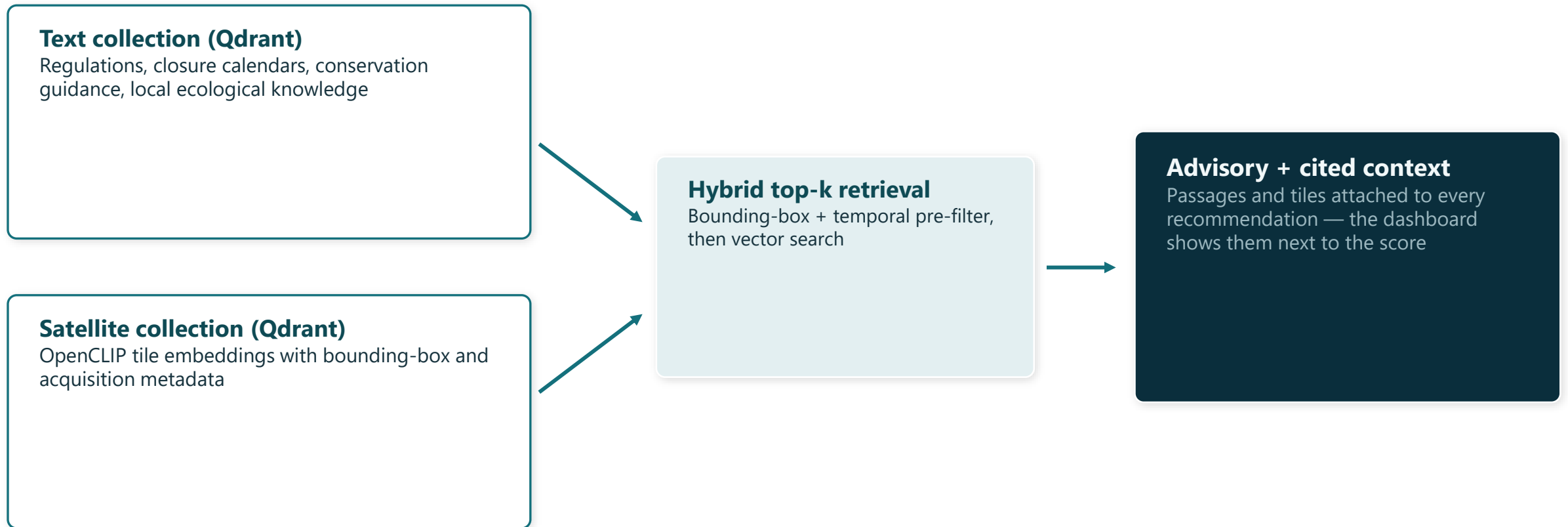
Local adaptation at 0.34% of the parameters



$$W = W_0 + BA$$

- Rank $r = 8$ updates on the query and value projections of all 12 encoder blocks
- 0.29 M of 85.8 M parameters trainable (0.34%) on ViT-B/16
- Training converges within mid-range GPU memory
- New regions and species adapted through configuration and lightweight fine-tunes — no code edits

Every advisory carries its own justification



Explainability is the retrieval of verifiable sources — not generated prose.

Scalable by design: new sources plug in as chunked documents — no model retraining; the corpus grows per region.

A grounded AI assistant explains - it never decides

Inputs (grounding only)

Live advisory JSON · top-k retrieved passages · optional image (catch photo, tile, chart) · user question + language



Qwen2.5-VL-7B (AWQ)

Single open-source VLM, Apache 2.0 · 4-bit weights $\approx$ 6.9 GB · served locally on vLLM with continuous batching



Guardrails (in code)

Citations must resolve to retrieved passages · verdict is authoritative · refusal over invention · backend named on every answer

Output: plain-language summary or cited answer + backend ID + authoritative verdict. If the GPU is down, a deterministic template path assembles the same schema from the same data — the website never goes dark.

Multilingual and scalable by design: answers in ten coastal languages, vernacular fish names, localised Problem/Answer self-explanations, ocean news translated into the reader's language (English pivot); a runtime model registry switches quality against hardware (3B laptop "> 72B datacenter).

T5 asserts: class consistency, citation integrity, cache, fallback, backend naming. T6 asserts: rate limits, 401/403 role gates, security headers, body cap.

Local AI on one GPU - no API bills at any traffic

Tier	Hardware	Model / profile	Sustains
L · laptop	RTX 4050 Laptop, 6 GB	Qwen2.5-VL-3B, Q4 · 6 k ctx · 1-2 concurrent	Personal use, demos
1 · pilot	RTX 3060/4070, 12 GB	7B AWQ · 8 k ctx · 8 concurrent	Thousands of requests/day
2 · recommended	RTX 4090 / NVIDIA L4, 24 GB	7B AWQ · 16 k ctx · 32-64 concurrent	Tens of thousands/day
3 · regional	2-4× L4/A10G + LB, or A100 40 GB	7B AWQ replicas + semantic cache	Millions of requests/year

\$0 per question

Self-hosted inference: the marginal cost of an answer is electricity.

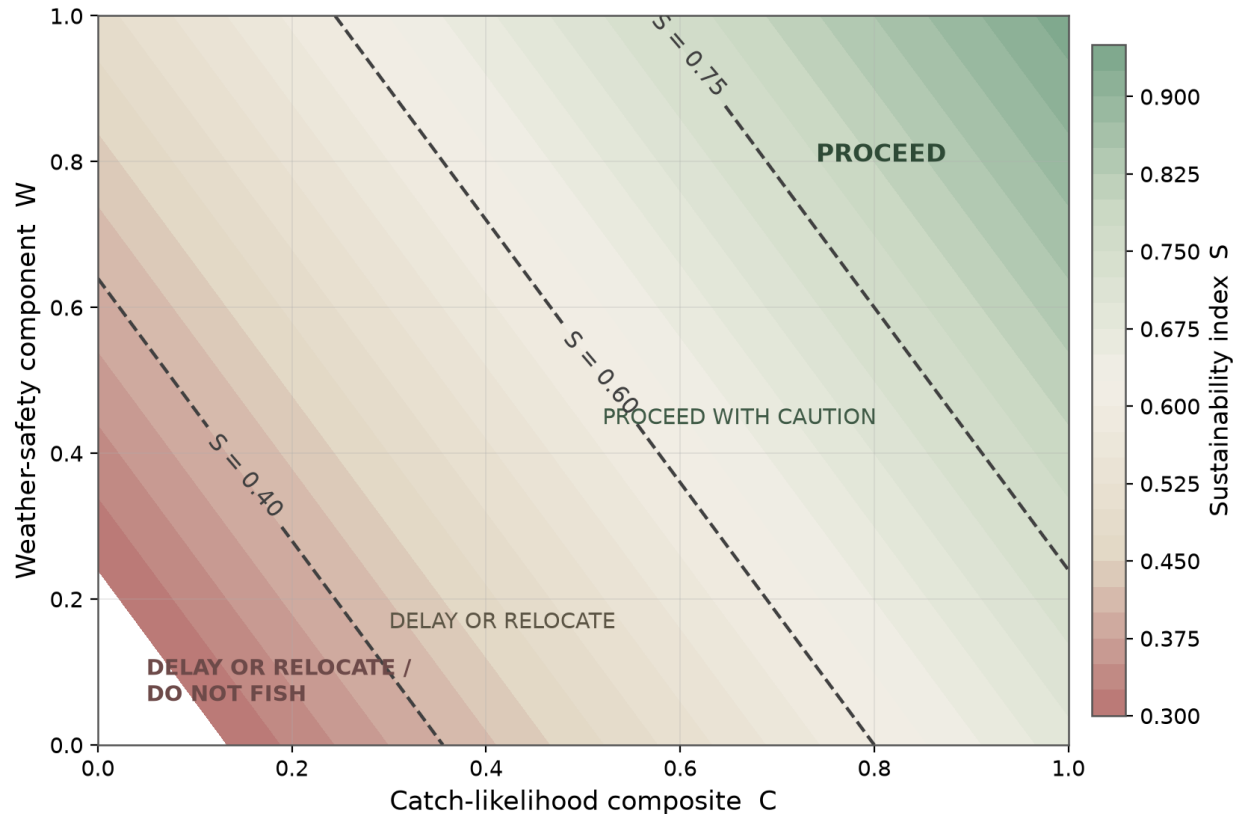
~\$16-55k/yr avoided

Per-token billing for 10k questions/day at commercial API rates.

~\$2k once + \$315/yr

The recommended 24 GB tier: one GPU plus power, owned outright.

One transparent sustainability index



$$S = 0.45 \cdot C + 0.25 \cdot W + 0.30 \cdot (1 - B)$$

PROCEED

$S \geq 0.75$

PROCEED WITH CAUTION

$0.60 \leq S < 0.75$

DELAY OR RELOCATE

$0.40 \leq S < 0.60$

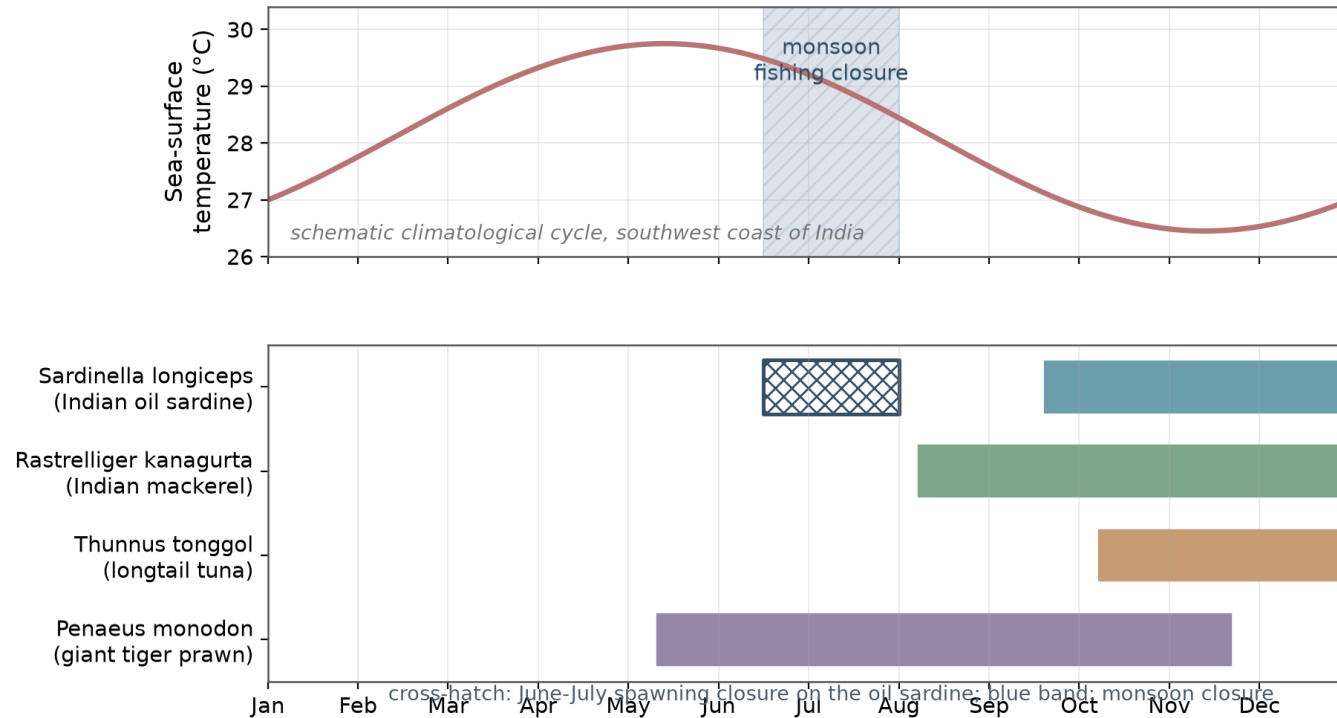
DO NOT FISH

$S < 0.40$

Hard rule: inside an active MPA or seasonal closure, the advisory is DO NOT FISH — regardless of score. Weights are configuration, owned by governance, not hidden constants.

CASE STUDY

Southwest coast of India

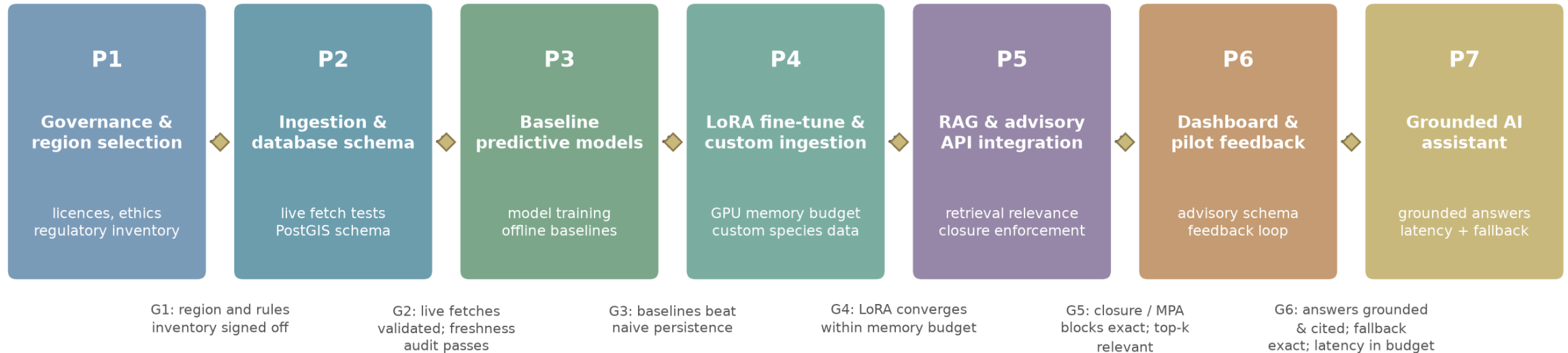


Why this coast

- Active closure regime: June–July monsoon closure, spawning protection for the oil sardine
- A network of marine protected areas enforced as hard constraints
- Sardine and mackerel fisheries dominate — and fluctuate strongly year to year
- Four coastal cities anchor the live-fetch validation tests

Registry: *Sardinella longiceps*, *Rastrelliger kanagurta*, *Thunnus tonggol*, *Penaeus monodon* — the prawn exercises the data-poor habitat fallback.

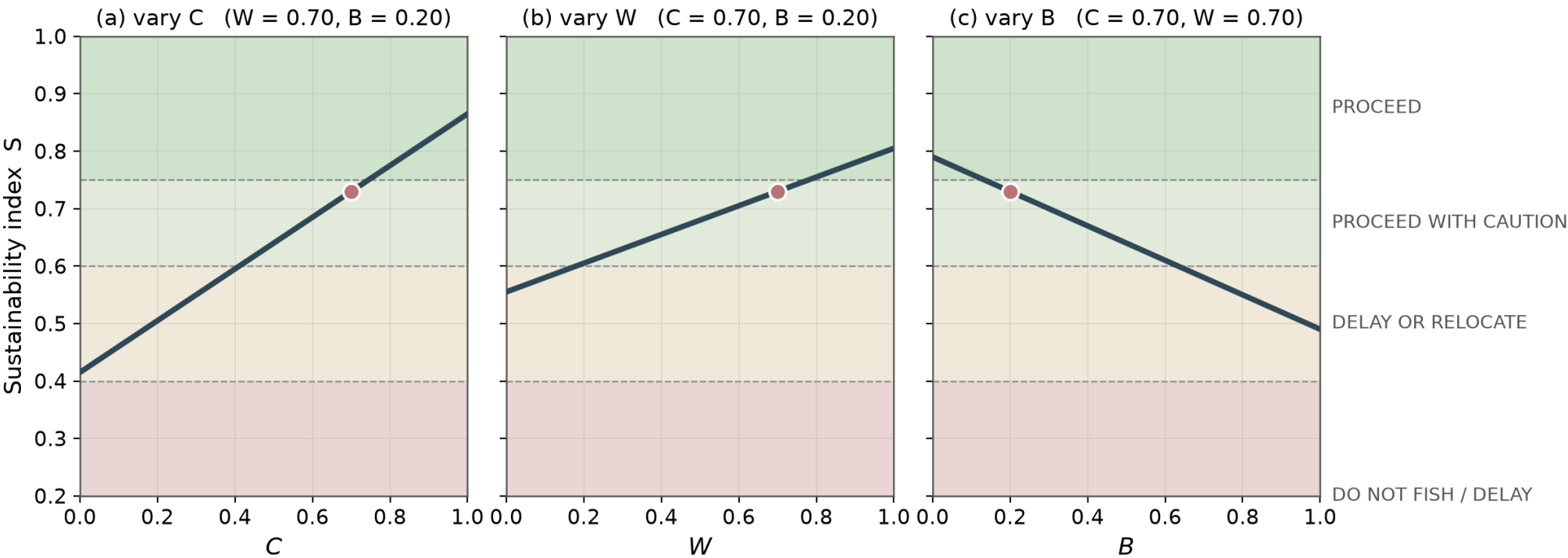
Phase-gate engineering, test-driven acceptance



iterative engineering with phase-gate acceptance checks (every gate must pass before the next phase begins)



Validated end to end



Live fetches reliable

Graceful degradation under outage, with explicit age flags

Closure blocks exact

June sardine request returns DO NOT FISH with the reason

Scoring monotonic

In every input; class thresholds land exactly as configured

Retrieval relevant

Top-k context attached to every advisory on the dashboard

Assistant guarded

Citations resolve; verdicts hold; template fallback exact

Service hardened

Rate limits 429; 401/403 role gates; headers on every response

Conservation, legality and safety are built into the objective function — ParaSail cannot recommend an illegal trip under any configuration.

Primary alignment: SDG 14 (Life Below Water) · supporting: SDG 2 · 8 · 9 · 13

The five mandatory summit questions

Q1 · Problem No real-time, integrated, explainable advisory system for sustainable fishing in data-poor regions.

Q2 · Importance Livelihoods, food security, marine conservation and safety all meet at this one information gap.

Q3 · Technology Geospatial AI (YOLOv8, ViT + LoRA, LSTM, random forest), dual-collection RAG, a self-hosted Qwen2.5-VL-7B assistant on vLLM with a runtime model registry, open live APIs, FastAPI + PostGIS + Qdrant; hardened with rate limiting and RBAC, shipped as an installable PWA.

Q4 · Evidence Open environmental, biodiversity and satellite data; regulation datasets; a working system validated by five test families.

Q5 · Contribution A production-grade platform issuing sustainability-scored, explainable, conversational advisories with conservation enforced as a hard constraint - on local AI, within 8-12 GB VRAM, at zero marginal cost.